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Liquid discharge system, server, and control method of liquid discharge system

Abstract

A liquid discharge system including a piezoelectric element that is displaced in accordance with a drive signal and discharges liquid; a drive signal output circuit that outputs the drive signal; a measurement circuit that measures a first capacitance of the piezoelectric element; and a determination circuit that determines a deterioration state of the piezoelectric element based on the first capacitance.

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Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2022-058936, filed Mar. 31, 2022, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a liquid discharge system, a server, and a control method of a liquid discharge system.

2. Related Art

(3) An ink jet printer or the like is known as a liquid discharge system that discharges liquid onto a medium. In the liquid discharge system, pressure is generated in a pressure generation chamber by displacement of a piezoelectric element so that liquid is discharged from a nozzle communicating with the pressure generation chamber. The discharged liquid lands on a medium so as to form a desired image on the medium. In such a liquid discharge system, a piezoelectric element is repeatedly driven such that a desired image is formed on the medium. Accordingly, the displacement amount of a piezoelectric element gradually decreases with repeated driving, which

causes a decrease in the amount of liquid discharged from a nozzle. Such a decrease in the liquid discharge amount might deteriorate the quality of an image formed on the medium.
(4) To deal with such a problem, JP-A-2009-066948 discloses a mechanism to count the number of liquid discharge times and to correct a voltage of a drive signal driving a piezoelectric element in accordance with the number of counts. The mechanism reduces the risk of decreasing the displacement amount of the piezoelectric element and thus reduces the risk of decreasing the liquid discharge amount from a nozzle.
(5) However, a decrease in the displacement amount of a piezoelectric element is caused not only by the number of drive times of a piezoelectric element, but also by a plurality of the other factors, such as a use environment, the value of a supply voltage, and the like. Accordingly, the technique disclosed in JP-A-2009-066948, which corrects the voltage of a drive signal driving a piezoelectric element in accordance with the number of discharge times, is not sufficient for dealing with the problem, and room for improvement thus remains.
SUMMARY
(6) According to an aspect of the present disclosure, there is provided a liquid discharge system including: a piezoelectric element that is displaced based on a drive signal and discharges liquid; a drive signal output circuit that outputs the drive signal; a measurement circuit that measures a first capacitance of the piezoelectric element; and a determination circuit that determines a deterioration state of the piezoelectric element based on the first capacitance.
(7) According to another aspect of the present disclosure, there is provided a server configured to communicate with a liquid discharge apparatus including a piezoelectric element that is displaced based on a drive signal and discharges liquid, the server including: a reception section that receives capacitance information including a capacitance of the piezoelectric element measured by a measurement circuit; a determination circuit that determines a deterioration state of the piezoelectric element based on the capacitance information; and a transmission section that transmits determination result information including a determination result of the determination circuit.
(8) According to still another aspect of the present disclosure, there is provided a control method of a liquid discharge system including a piezoelectric element that is displaced based on a drive signal and discharges liquid, and a drive signal output circuit that outputs the drive signal, the control method including: a measurement step of measuring a capacitance of the piezoelectric element; and a determination step of determining a deterioration state of the piezoelectric element based on the capacitance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a diagram illustrating the configuration to a liquid discharge system.
(2) FIG. **2** is a diagram illustrating an example of the signal waveform of a drive signal COM.
(3) FIG. **3** is a diagram illustrating an example of the schematic configuration of a discharge section.
(4) FIG. **4** is a diagram illustrating an example of the configuration of a measurement circuit.
(5) FIG. **5** is a diagram for explaining an example of the operation of the measurement circuit.
(6) FIG. **6** is a diagram illustrating an example of the control method of the liquid discharge system.
(7) FIG. **7** is a diagram illustrating the configuration to a liquid discharge system according to a second embodiment.
(8) FIG. **8** is a diagram illustrating an example of the configuration of a measurement circuit according to the second embodiment.

(9) FIG. **9** is a diagram illustrating the configuration to a liquid discharge system according to a third embodiment.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

(10) In the following, a description will be given of preferred embodiments of the present disclosure with reference to the drawings. The drawings provided are illustrated for convenience of explanation. In this regard, the embodiments described below do not unreasonably restrict the contents described in the scope of the claims. Note that not all the components of the configurations described below are necessarily mandatory constituent features.

1. First Embodiment

(11) Functional Configuration and Operation of Liquid Discharge System SY

(12) FIG. **1** is a diagram illustrating the configuration of a liquid discharge system SY according to a first embodiment. As illustrated in FIG. **1**, the liquid discharge system SY according to the first embodiment includes a liquid discharge apparatus **1** that discharges liquid. A description will be given on the assumption that the liquid discharge apparatus **1** according to the first embodiment is a so-called ink jet printer that discharges ink, as an example of a liquid, onto a medium so as to form a desired image on the medium. However, the liquid discharge apparatus **1** is not limited to this. The liquid discharge apparatus **1** may be a color material discharge apparatus used for manufacturing a color filter of a liquid crystal display or the like, an electrode material discharge apparatus used for forming an electrode of an organic EL display, a surface emitting display, or the like, a bioorganic matter discharge apparatus used for biochip manufacturing, or the like.

(13) As illustrated in FIG. **1**, the liquid discharge apparatus **1** includes a control circuit **100**, a notification section **110**, a drive signal output circuit **50**, a drive signal selection control circuit **210**, n selection circuits **230**, a measurement circuit **300**, a determination circuit **310**, a life estimation circuit **320**, a correction value calculation circuit **330**, and n discharge sections **600**.

(14) Image data including information on an image to be formed on the medium is input to the control circuit **100** from an external device not illustrated in the figure and disposed outside the liquid discharge apparatus **1**. The control circuit **100** then controls each component of the liquid discharge apparatus **1** in accordance with the input image data.

(15) Specifically, the control circuit **100** generates a discharge data signal DATA based on the image data input from the external device and outputs the discharge data signal DATA to the drive signal selection control circuit **210**. The drive signal selection control circuit **210** generates selection signals S corresponding to n selection circuits **230** which are defined by the discharge data signal DATA at the timing defined by the input discharge data signal DATA and outputs the selection signals S to the respective selection circuits **230**. That is to say, the discharge data signal DATA includes a signal defining the timing when the drive signal selection control circuit **210** outputs the selection signals S and signals defining the logical level of the selection signals S corresponding to the respective n selection circuits **230**.

(16) The control circuit **100** generates a reference drive signal dA and outputs the reference drive signal dA to the drive signal output circuit **50**. The drive signal output circuit **50** converts the reference drive signal dA, which is the input digital signal, into an analog signal and amplifies the converted analog signal so as to generate a drive signal COM. The drive signal output circuit **50** then outputs the drive signal COM to n selection circuits **230**. That is to say, the control circuit **100** outputs the reference drive signal dA that defines the signal waveform of the drive signal COM to the drive signal output circuit **50**, and the drive signal output circuit **50** amplifies the signal waveform defined by the input reference drive signal dA so as to generate the drive signal COM and outputs the drive signal COM to n selection circuits **230**. In other words, the liquid discharge system SY includes the drive signal output circuit **50** that outputs the drive signal COM.

(17) It is preferable that the drive signal output circuit **50** as described above amplify an analog signal based on the reference drive signal dA and include an amplifier circuit, such as a class A amplifier circuit, a class B amplifier circuit, a class AB amplifier circuit, a class D amplifier circuit,

or the like. It is also preferable that the reference drive signal dA output by the control circuit **100** define the signal waveform of the drive signal COM output by the drive signal output circuit **50** and be an analog signal.
(18) Here, a description will be given of an example of the signal waveform of the drive signal COM output by the drive signal output circuit **50**. FIG. **2** is a diagram illustrating an example of the signal waveform of the drive signal COM. As illustrated in FIG. **2**, the drive signal COM includes a trapezoid waveform Adp for each predetermined period. The trapezoid waveform Adp includes a fixed period at a voltage vc, a fixed period at a voltage vb having a voltage value lower than the voltage vc subsequent to the fixed period at the voltage vc, a fixed period at a voltage vt having a voltage value higher than the voltage vc subsequent to the fixed period at the voltage vb, and a fixed period at the voltage vc subsequent to the fixed period at the voltage vt. That is to say, the drive signal output circuit **50** outputs the drive signal COM including the trapezoid waveform Adp that begins at the voltage vc, changes to voltages vb and vt, and ends at the voltage vc.
(19) Here, in the following description, the potential difference between the voltage vt and the voltage vb may be referred to as the amplitude of the drive signal COM. In this regard, the signal waveform of the drive signal COM output by the drive signal output circuit **50** is not limited to the shape illustrated in FIG. **2**. The drive signal output circuit **50** may output a drive signal COM including a signal waveform having various shapes in accordance with the physical property of the ink discharged onto the medium, the frequency of the drive signal COM, the transport speed of the medium, and the like.
(20) Referring back to FIG. **1**, a selection signal S output by the drive signal selection control circuit **210** and the drive signal COM output by the drive signal output circuit **50** are input to the n selection circuits **230**. The n selection circuits **230** then determine the trapezoid waveform Adp of the drive signal COM to be selected or deselected in accordance with the logical level of the input selection signal S so as to generate and output a drive signal VOUT. Specifically, when a selection circuit **230** determines the trapezoid waveform Adp of the drive signal COM to be selected based on the logical level of the selection signal S, the selection circuit **230** outputs the drive signal VOUT including the trapezoid waveform Adp. On the other hand, when a selection circuit **230** determines the trapezoid waveform Adp of the drive signal COM to be deselected based on the logical level of the selection signal S, the selection circuit **230** outputs the drive signal VOUT not including the trapezoid waveform Adp. At this time, a voltage Vc having a fixed value, which corresponds to the immediately preceding voltage and is held by a capacitance component of a piezoelectric element **60**, is held at the output end of the selection circuit **230**. That is to say, when the selection circuit **230** determines the trapezoid waveform Adp of the drive signal COM to be deselected based on the logical level of the selection signal S, the selection circuit **230** outputs the drive signal VOUT having the fixed voltage Vc.
(21) The n discharge sections **600** have respective piezoelectric elements **60** and are disposed correspondingly to the n selection circuits **230**. One end of the piezoelectric elements **60** included in the respective n discharge sections **600** is supplied with the drive signals VOUT output by the corresponding selection circuits **230**. The other end of the piezoelectric elements **60** included in the respective n discharge sections **600** is commonly supplied with a voltage Vss. The piezoelectric elements **60** included in the respective n discharge sections **600** are driven by the respective potential differences between the drive signals VOUT supplied to one end and the voltage Vss supplied to the other end. The amount of ink in accordance with driving of the piezoelectric elements **60** is output from the discharge section **600**. Here, the voltage Vss supplied to the other end of the piezoelectric elements **60** is a reference voltage signal for driving the piezoelectric elements **60** and may be, for example, a ground potential having a fixed voltage or a fixed voltage signal of 5.5 V or 6 V.
(22) Here, a description will be given of a specific example of the configuration of the discharge sections **600** including the respective piezoelectric elements **60**. FIG. **3** is a diagram illustrating an

example of the schematic configuration of one discharge section **600** of the n discharge sections **600**. FIG. **3** also illustrates a nozzle plate **632**, a reservoir **641**, and a supply port **661** in addition to the discharge section **600**. As illustrated in FIG. **3**, the discharge section **600** includes the piezoelectric element **60**, a vibration plate **621**, a cavity **631**, and a nozzle **651**.

(23) The piezoelectric element **60** includes a piezoelectric body **601** and electrodes **611** and **612**. In the piezoelectric element **60**, the electrodes **611** and **612** are positioned so as to sandwich the piezoelectric body **601**. The piezoelectric element **60** as described above is driven such that the central part thereof is displaced in the vertical direction in accordance with the potential difference between the voltage of the signal supplied to the electrode **611** and the voltage of the signal supplied to the electrode **612**. Specifically, the electrode **611**, which is one end, is supplied with the drive signal VOUT, and the electrode **612**, which is the other end, is supplied with the voltage Vss. When the voltage of the drive signal VOUT, supplied to the electrode **611**, is changed, the potential difference between the drive signal VOUT supplied to the electrode **611** and the voltage Vss supplied to the electrode **612** is changed. As a result, the central part of the piezoelectric element **60** is driven so as to be displaced in the vertical direction. On the other hand, when the voltage of the drive signal VOUT, supplied to the electrode **611**, is not changed, the potential difference between the drive signal VOUT supplied to the electrode **611** and the voltage Vss supplied to the electrode **612** is not changed. As a result, the displacement of the central part of the piezoelectric element **60** is held in a fixed state. That is to say, when the drive signal VOUT including the trapezoid waveform Adp with a changing voltage value is supplied to the electrode **611**, the piezoelectric element **60** is driven, whereas when the electrode **611** is supplied with the drive signal VOUT having the fixed voltage Vc, the piezoelectric element **60** is not driven.

(24) The vibration plate **621** is positioned below the piezoelectric element **60** in FIG. **3**. In other words, the piezoelectric element **60** is formed on the upper surface of the vibration plate **621** in FIG. **3**. The vibration plate **621** as described above is displaced in the vertical direction with the driving of the piezoelectric element **60**.

(25) The cavity **631** is positioned below the vibration plate **621** in FIG. **3**. Ink is supplied to the cavity **631** from the reservoir **641**. Ink stored in an ink cartridge, not illustrated in the figure, held by the liquid discharge apparatus **1** or the like is fed to the reservoir **641** via the supply port **661**. The internal volume of the cavity **631** as described above is expanded or reduced with the displacement of the vibration plate **621** in the vertical direction. That is to say, the vibration plate **621** functions as a diaphragm that changes the internal volume of the cavity **631**, and the cavity **631** functions as a chamber that changes the pressure with displacement of the vibration plate **621** in the vertical direction.

(26) The nozzle **651** is an opening disposed on the nozzle plate **632** and communicates with the cavity **631**. When the internal volume of the cavity **631** is changed, the internal pressure of the cavity **631** is changed in accordance with a change in the internal volume, and the ink filled in the cavity **631** is discharged from the nozzle **651**.

(27) In the discharge section **600** configured as described above, when the piezoelectric element **60** is driven so as to be bent in the up direction, the vibration plate **621** is displaced in the up direction. Thereby, the internal volume of the cavity **631** is expanded. As a result, the ink stored in the reservoir **641** is drawn into the cavity **631**. On the other hand, when the piezoelectric element **60** is driven so as to be bent in the down direction, the vibration plate **621** is displaced in the down direction. Thereby, the internal volume of the cavity **631** is reduced. As a result, the amount of ink in accordance with the degree of the reduction of the internal volume of the cavity **631** is discharged from the nozzle **651**.

(28) Specifically, in the period in which the drive signal VOUT having a fixed voltage Vc is supplied to the electrode **611** of the piezoelectric element **60**, the displacement amount of the piezoelectric element **60** remains fixed. Accordingly, ink is not discharged from a nozzle **651** corresponding to the piezoelectric element **60**. Next, when the voltage of the drive signal VOUT

supplied to the electrode **611** of the piezoelectric element **60** is changed from the voltage vc to the voltage vb, the piezoelectric element **60** is driven so as to be displaced in the up direction illustrated in FIG. **3** in accordance with the change in the voltage of the drive signal VOUT. Thereby, the vibration plate **621** is displaced in the up direction illustrated in FIG. **3**, and the internal volume of the cavity **631** is expanded. As a result, the ink stored in the reservoir **641** is drawn into the cavity **631**. Next, when the voltage of the drive signal VOUT supplied to the electrode **611** of the piezoelectric element **60** is changed from the voltage vb to the voltage vt, the piezoelectric element **60** is driven so as to be displaced in the down direction illustrated in FIG. **3** in accordance with the change in the voltage of the drive signal VOUT. Thereby, the vibration plate **621** is displaced in the down direction illustrated in FIG. **3**, and the internal volume of the cavity **631** is reduced. As a result, the ink stored in the cavity **631** is discharged from the nozzle **651**.

(29) As described above, the piezoelectric element **60** is driven in accordance with a change in the voltage of the drive signal VOUT supplied to the electrode **611**. The discharge section **600** then discharges the amount of ink in accordance with the drive amount of the piezoelectric element **60**. In other words, the piezoelectric element **60** is displaced in accordance with the drive signal VOUT based on the drive signal COM to discharge ink. At this time, the displacement amount of the piezoelectric element **60**, which is the drive amount, and the ink discharge amount depend on the characteristic of the piezoelectric element **60** and the respective values of the voltages vc, vt, and vb supplied to the piezoelectric element **60**.

(30) In this regard, it is preferable that the piezoelectric element **60** be configured to be driven in accordance with the voltage value of the supplied drive signal VOUT and discharge ink from a corresponding nozzle **651** when driven, noting that such a configuration is not limited to the one example illustrated in FIG. **3**.

(31) Referring back to FIG. **1**, the measurement circuit **300** obtains at least one voltage value of the n piezoelectric elements **60** as a detection voltage Cc. Specifically, the measurement circuit **300** obtains the voltage value of at least one electrode **611** of the n piezoelectric elements **60** as the detection voltage Cc by receiving input of an enable signal EN, which enables detection of a voltage value, from the control circuit **100**. The measurement circuit **300** then generates a detection signal Cap corresponding to the capacitance of the piezoelectric element **60** based on the obtained detection voltage Cc and outputs the detection signal Cap to the determination circuit **310**. That is to say, the measurement circuit **300** measures the capacitance of the piezoelectric element **60** and outputs the detection signal Cap corresponding to the capacitance. In this regard, a detailed description will be given later of the measurement circuit **300**.

(32) Here, the measurement of the capacitance by the measurement circuit **300** is not limited to direct measurement of the capacitance of the piezoelectric element **60** by the measurement circuit **300** and includes the case of measuring a physical quantity that changes in accordance with the capacitance value. Specifically, the measurement circuit **300** may obtain the voltage value of the piezoelectric element **60** when a fixed current or voltage is supplied to the piezoelectric element **60** so as to measure a time required for charging the piezoelectric element **60** that changes with the capacitance of the piezoelectric element **60** as the capacitance of the piezoelectric element **60**, and may output a measurement result as the detection signal Cap. The measurement circuit **300** may also obtain the voltage value of the piezoelectric element **60** when a charge stored in the piezoelectric element **60** is discharged so as to measure a time required for discharging a charge that is changed by the capacitance of the piezoelectric element **60** as the capacitance of the piezoelectric element **60**, and may output a measurement result as the detection signal Cap. Further, the measurement circuit **300** may measure the value of the charging voltage provided to the piezoelectric element **60** in a fixed time that changes with the capacitance of the piezoelectric element **60** as the capacitance of the piezoelectric element **60**, and may output a measurement result as the detection signal Cap.

(33) In this regard, in FIG. **1**, the example in which the measurement circuit **300** detects the voltage

value of one piezoelectric element **60** is illustrated. However, the measurement circuit **300** may detect the voltage values of a plurality of piezoelectric elements **60**. That is to say, the number of piezoelectric elements **60** whose capacitance is measured by the measurement circuit **300** is not limited to one and may be plural. Also, when the measurement circuit **300** measures the capacitance of a plurality of piezoelectric elements **60**, the measurement circuit **300** may separately measure the capacitances of the plurality of piezoelectric elements **60** or may measure the combined capacitance of the plurality of piezoelectric elements **60**.

(34) The determination circuit **310** receives an input of the detection signal Cap. The determination circuit **310** determines a deterioration state of the piezoelectric element **60** based on the input detection signal Cap. For example, the determination circuit **310** may determine the deterioration state of the piezoelectric element **60** in accordance with the capacitance value of the piezoelectric element **60** based on the detection signal Cap. Also, the determination circuit **310** may calculate a change rate of the capacitance of the piezoelectric element **60** from the capacitance of the piezoelectric element **60** based on the input detection signal Cap and the initial value of the capacitance of the piezoelectric element **6**, and may determine the deterioration state of the piezoelectric element **60** in accordance with the calculated change rate. The determination circuit **310** then classifies the deterioration state of the piezoelectric element **60** based on a determination result. For example, the determination circuit **310** classifies the deterioration state into a normal rank, a subnormal rank, a slightly abnormal rank, a moderately abnormal rank, and a severely abnormal rank. Next, the determination circuit **310** generates a state signal Cst including the information on the rank as a determination result of the deterioration state of the piezoelectric element **60**, and outputs the state signal Cst to the life estimation circuit **320** and the correction value calculation circuit **330**. In this regard, the classification of the rank determined by the determination circuit **310** is not limited to the classification described above.

(35) The life estimation circuit **320** estimates the life of the piezoelectric element **60** based on the input state signal Cst. For example, when the rank of the deterioration state based on the input state signal Cst is the normal rank, the subnormal rank, or the slightly abnormal rank, the life estimation circuit **320** estimates that the piezoelectric element **60** has a sufficient life. Also, when the rank of the deterioration state based on the input state signal Cst is the moderately abnormal rank, the life estimation circuit **320** estimates that the piezoelectric element **60** soon reaches the end of life. Also, when the rank of the deterioration state based on the input state signal Cst is the severely abnormal rank, the life estimation circuit **320** estimates that the piezoelectric element **60** has already reached the end of its life. The life estimation circuit **320** then generates a life notification signal El including the information on the estimated life of the piezoelectric element **60**, and outputs the life notification signal El to the control circuit **100**.

(36) When the life notification signal El input from the life estimation circuit **320** includes the information stating that the piezoelectric element **60** will soon reach the end of life or the information stating that the piezoelectric element **60** has already reached the end of life, the control circuit **100** generates a notification signal Inf for notifying a user of that the piezoelectric element **60** will soon reach the end of life, or that the piezoelectric element **60** has already reached the end of life, and outputs the notification signal Inf to the notification section **110**. The notification section **110** notifies of the life information on the piezoelectric element **60** based on the input notification signal Inf. That is to say, the liquid discharge system SY includes the life estimation circuit **320** that estimates the life of the piezoelectric element **60** based on the deterioration state of the piezoelectric element **60** determined by the determination circuit **310** and the notification section **110** that notifies of the information on the life estimated by the life estimation circuit **320**.

(37) The correction value calculation circuit **330** determines whether or not to correct the voltage of the drive signal COM based on the input state signal Cst. Specifically, when the rank of the deterioration state based on the input state signal Cst is the normal rank or the subnormal rank, the correction value calculation circuit **330** determined that it is not necessary to correct the voltage of

the drive signal COM. Also, when the rank of the deterioration state based on the input state signal Cst is the slightly abnormal rank, the correction value calculation circuit **330** generates a correction signal Cv for correcting the voltage of the drive signal COM by a voltage v**1**, and outputs the correction signal Cv to the control circuit **100**. Also, when the rank of the deterioration state based on the input state signal Cst is the moderately abnormal rank, the correction value calculation circuit **330** generates the correction signal Cv for correcting the voltage of the drive signal COM by a voltage v**2** higher than the voltage v**1**, and outputs the correction signal Cv to the control circuit **100**. Also, when the rank of the deterioration state based on the input state signal Cst is the severely abnormal rank, the correction value calculation circuit **330** generates the correction signal Cv for correcting the voltage of the drive signal COM by a voltage v**3** higher than the voltage v**2** and outputs the correction signal Cv to the control circuit **100**. In this regard, when the rank of the deterioration state based on the input state signal Cst is the severely abnormal rank, the correction value calculation circuit **330** may generate a correction signal Cv that stops the output of the drive signal COM and may output the correction signal Cv to the control circuit **100**.

(38) The control circuit **100** corrects the reference drive signal dA in accordance with the correction signal Cv input from the correction value calculation circuit **330**. Thereby, the voltage of the drive signal COM generated based on the reference drive signal dA is corrected. Specifically, the control circuit **100** may make a correction so as to increase the voltage value of the voltage vt of the trapezoid waveform Adp included in the drive signal COM as the correction of the reference drive signal dA in accordance with the correction signal Cv input from the correction value calculation circuit **330**. Also, the control circuit **100** may make a correction so as to increase the amplitude of the drive signal COM. That is to say, the correction value calculation circuit **330** corrects the voltage of the drive signal COM based on the state signal Cst in accordance with the detection signal Cap with the control circuit **100**.

(39) As described above, the liquid discharge system SY according to the first embodiment includes the piezoelectric element **60** that is displaced in accordance with the drive signal VOUT based on the discharges in drive signal COM to discharge ink, the drive signal output circuit **50** that outputs the drive signal COM, the measurement circuit **300** that measures the capacitance of the piezoelectric element **60**, and the determination circuit **310** that determines the deterioration state of the piezoelectric element **60** based on the capacitance. Further, the liquid discharge system SY includes the correction value calculation circuit **330** that corrects the voltage of the drive signal COM based on the deterioration state of the piezoelectric element **60** determined by the determination circuit **310**, the control circuit **100**, the life estimation circuit **320** that estimates the life of the piezoelectric element **60** based on the deterioration state determined by the determination circuit **310**, and the notification section **110** that notifies of the information on the life of the piezoelectric element **60** estimated by the life estimation circuit **320**.

(40) Here, with the liquid discharge system SY that drives the piezoelectric element **60** by the drive signal COM so as to discharge ink by driving the piezoelectric element **60**, there is a widespread phenomenon that the displacement amount of the piezoelectric element is deteriorated by continuous driving of the piezoelectric element or the like. Up to this time, to deal with deterioration of the displacement amount that occurs with the piezoelectric element, the number of driven times of the piezoelectric element by the drive signal is counted, and the voltage of the drive signal is corrected so that deterioration of the drive signal that has occurred in the piezoelectric element is compensated. However, deterioration of the displacement amount that has occurred in the piezoelectric element is not only caused by the number of drive times of the piezoelectric element, but also caused by the drive conditions, such as the drive time of the piezoelectric element **60**, the voltage of the drive signal COM supplied to the piezoelectric element **60**, the ambient temperature of the piezoelectric element **60**, and the like. Accordingly, by the related-art method in which the number of driven times of the piezoelectric element by the drive signal is counted to correct the drive signal, it has been difficult to suitably determine deterioration of the displacement

amount of the drive element.
(41) To handle such a problem, the present inventor focused attention on the point that there is a correlation between deterioration of the displacement amount of the piezoelectric element and a polarization change of the piezoelectric element, and realized the mechanism as follows. In the liquid discharge system SY, the measurement circuit **300** measures the capacitance of the piezoelectric element **60**, and the determination circuit **310** determines the deterioration state of the piezoelectric element **60** based on the capacitance of the piezoelectric element **60** measured by the measurement circuit **300** so that the deterioration of the amount of change of the piezoelectric element **60** is detected without depending on the drive condition of the piezoelectric element **60**.
(42) Further, the liquid discharge system SY according to the first embodiment includes the correction value calculation circuit **330** that corrects the voltage of the drive signal COM based on the deterioration state of the piezoelectric element **60** determined by the determination circuit **310**. Thereby, when deterioration occurs in the displacement amount of the piezoelectric element **60**, it is possible to reduce the risk of variation of the amount of discharge liquid. That is to say, the discharge accuracy of ink in the liquid discharge system SY is improved.
(43) Also, deterioration of the capacitance of the piezoelectric element **60** is caused by continuous drive of the piezoelectric element **60**, and thus the capacitance of the piezoelectric element **60** decreases at the end of life. The liquid discharge system SY according to the first embodiment includes the life estimation circuit **320** that estimates the life of the piezoelectric element **60** based on the deterioration state of the piezoelectric element **60** determined by the determination circuit **310**, and the notification section **110** that provides a notification of the information on the life of the piezoelectric element **60** estimated by the life estimation circuit **320**. Accordingly, it is possible to notify the user of the life of the piezoelectric element **60** and the liquid discharge system SY at a suitable timing, and thus to improve user convenience.
(44) Configuration and Operation of Measurement Circuit
(45) Here, a description will be given of an example of the specific configuration and operation of the measurement circuit **300** that measures the capacitance of the piezoelectric element **60**. FIG. **4** is a diagram illustrating an example of the configuration of the measurement circuit **300**. As illustrated in FIG. **4**, the measurement circuit **300** includes a current source **301** that outputs a constant current, a switch **302** that changes an electrical connection between the current source **301** and the piezoelectric element **60**, a comparator **303** electrically coupled to the piezoelectric element **60**, and a timer circuit **304** that measures a charging time Tch of the piezoelectric element **60** based on the output of the comparator **303**.
(46) The current source **301** is supplied with a voltage Vdd**1**. The current source **301** then generates a current Icnt having a fixed current value based on the voltage Vdd**1** and outputs the current Icnt. The current source **301** as described above may be configured, for example, by one or a combination of a plurality of transistors, a constant-current Zener diode, or the like.
(47) One end of the switch **302** is electrically coupled to the output of the current source **301**, and the other end thereof is electrically coupled to the electrode **611** of the piezoelectric element **60**. The conduction state of one end and the other end of the switch **302** is changed based on the enable signal EN input to the control end. Specifically, when the enable signal EN, which is an enable signal EN that enables detection of the capacitance of the piezoelectric element **60** in the measurement circuit **300** and enables the detection of the detection voltage Cc of the piezoelectric element **60**, is input to the control end of the switch **302**, the one end and the other end of the switch **302** are controlled to be conductive. When the enable signal EN, which is an enable signal EN that does not enable detection of the capacitance of the piezoelectric element **60** in the measurement circuit **300** and does not enable the detection of the detection voltage Cc of the piezoelectric element **60**, is input to the control end of the switch **302**, the one end and the other end of the switch **302** are controlled to be nonconductive. That is to say, the switch **302** controls the supply of the current Icnt to the piezoelectric element **60** depending on the enable signal EN.

(48) A voltage Vdd**2** is supplied to the high potential side power source terminal of the comparator **303**, and the voltage Vss is supplied to the low potential side power source terminal of the comparator **303**. Also, one input end of the comparator **303** is electrically coupled to the electrode **611** of the piezoelectric element **60**, and a reference voltage Vref is input to the other input end of the comparator **303**. When the detection voltage Cc, which is the voltage value of the electrode **611** of the piezoelectric element **60** input into the one input end, is higher than the reference voltage Vref, the comparator **303** outputs the H-level output voltage Cp, whereas when the detection voltage Cc is less than or equal to the reference voltage Vref, the comparator **303** outputs the L-level output voltage Cp.
(49) The output voltage Cp output from the comparator **303** is input to the timer circuit **304**. Also, the timer circuit **304** receives input of the enable signal EN and a clock pulse not illustrated in the figure. The timer circuit **304** calculates a time period from when the enable signal EN that enables detection of the capacitance of the piezoelectric element **60** in the measurement circuit **300** and that enables detection of the detection voltage Cc of the piezoelectric element **60** is input to the time when the output voltage Cp output from the comparator **303** becomes the H level based on the clock pulse. The timer circuit **304** outputs a calculation result as the detection signal Cap.
(50) A description will be given of an example of the operation of the measurement circuit **300** configured as described above. FIG. **5** is a diagram illustrating an example of the operation of the measurement circuit **300**. In this regard, FIG. **5** is illustrated on the assumption that the logical level of the enable signal EN that enables detection of the capacitance of the piezoelectric element **60** in the measurement circuit **300** is H level, and the logical level of the enable signal EN that does not enable detection of the capacitance of the piezoelectric element **60** in the measurement circuit **300** is L level.
(51) As illustrated in FIG. **5**, when the H-level enable signal EN is input to the measurement circuit **300** at time t**1**, one end and the other end of the switch **302** are controlled to be conductive. Thereby, the current Icnt output by the current source **301** is supplied to the piezoelectric element **60** via the switch **302**. Accordingly, the voltage value of the piezoelectric element **60** rises based on the capacity component of the piezoelectric element **60** and the current value of the current Icnt. That is to say, the detection voltage Cc indicating the voltage value of the piezoelectric element **60** gradually increases.
(52) Also, the H-level enable signal EN is input to the timer circuit **304** at time t**1**. The timer circuit **304** starts measuring time by receiving the input of the H-level enable signal EN. Specifically, the timer circuit **304** starts counting the number of pulses of the input clock pulse not illustrated in the figure.
(53) Next, the detection voltage Cc indicating the voltage value of the piezoelectric element **60** exceeds the reference voltage Vref at time t**2** so that the comparator **303** outputs the H-level output voltage Cp. The H-level output voltage Cp output by the comparator **303** is input to the timer circuit **304**. The timer circuit **304** stops measuring time, that is to say, counting the number of pulses of the clock pulse by receiving input of the H-level output voltage Cp. The timer circuit **304** calculates the charging time Tch based on the counted number of pulses of the clock pulse. Next, the timer circuit **304** generates the detection signal Cap including the calculated charging time Tch and outputs the detection signal Cap.
(54) At time t**3** after the timer circuit **304** outputs the detection signal Cap, the L-level enable signal EN is input to the measurement circuit **300**. Thereby, supply of the current Icnt output by the current source **301** to the piezoelectric element **60** is stopped. Accordingly, the voltage value of the piezoelectric element **60** gradually decreases. The detection voltage Cc indicating the voltage value of the piezoelectric element **60** becomes lower than or equal to the reference voltage Vref so that the comparator **303** outputs the L-level output voltage Cp. Thereby, calculation of the capacitance of the piezoelectric element **60** in the measurement circuit **300** is ended.
(55) That is to say, in the measurement circuit **300** according to the first embodiment, by obtaining

the detection voltage Cc indicating the voltage value of the piezoelectric element **60** when the current Icnt is supplied to the piezoelectric element **60**, the charging time Tch required for charging the piezoelectric element **60**, which changes depending on the capacitance of the piezoelectric element **60**, is measured, and a measurement result is output as the detection signal Cap.
(56) Here, as a method of measuring the capacitance by the measurement circuit **300** is not limited to the method described with reference to FIG. **4** and FIG. **5**, and various techniques configured to measure the capacitance of the piezoelectric element **60** may be applied. However, as illustrated in FIG. **4** and FIG. **5**, the measurement circuit **300** may be configured to include the current source **301** that outputs a constant current, the switch **302** that changes the electrical connection between the current source **301** and the piezoelectric element **60**, the comparator **303** electrically coupled to the piezoelectric element **60**, and the timer circuit **304** that measures the charging time Tch of the piezoelectric element **60** based on the output of the comparator **303**. By configuring the measurement circuit **300** as described above, it becomes possible to configure the measurement circuit **300** by a plurality of transistors. As a result, it is possible to realize the measurement circuit **300** as one or a plurality of integrated circuits. Thereby, it becomes possible to miniaturize the measurement circuit **300**.
(57) Control Method of Liquid Discharge System
(58) Here, a description will be given of the control method of the liquid discharge system SY described above. A control method of the liquid discharge system SY according to the present embodiment is a control method of the liquid discharge system SY including the piezoelectric element **60** that is displaced based on the drive signal COM to discharge ink and the drive signal output circuit **50** that outputs the drive signal COM, and includes a measurement step of measuring the capacitance of the piezoelectric element **60** and a determination step of determining the deterioration state of the piezoelectric element **60** based on the measured capacitance.
(59) FIG. **6** is a diagram illustrating an example of the control method of the liquid discharge system SY. As illustrated in FIG. **6**, in the liquid discharge system SY according to the first embodiment, the control circuit **100** of the liquid discharge apparatus **1** counts the number of discharge times of ink onto a medium. The control circuit **100** then determines whether or not the number of discharge times of ink onto the medium has reached a predetermined number of times (step S**110**).
(60) When the control circuit **100** determines that the number of discharge times of ink onto a medium has not reached a predetermined number of times (N in step S**110**), the control circuit **100** does not determine the deterioration state of the piezoelectric element **60** and continues counting the number of discharge times. On the other hand, when the control circuit **100** determines that the number of discharge times of ink onto the medium has reached the predetermined number of times medium (Y in step S**110**), the control circuit **100** performs the determination of the deterioration state of the piezoelectric element **60** and the processing based on the determination result of the deterioration state. That is to say, in the liquid discharge system SY according to the first embodiment, the control circuit **100** determines the deterioration state of the piezoelectric element **60** by using the number of discharge times of ink onto the medium as a trigger, and determines whether or not the processing based on the determination result of the deterioration state is necessary.
(61) In this regard, the trigger for the control circuit **100** to determine the deterioration state of the piezoelectric element **60** and determine whether or not the processing based on the determination result of the deterioration state is not necessary limited to the number of discharge times of ink described above. For example, the trigger may be when the drive time of the liquid discharge apparatus **1** has reached a predetermined time, when the number of media on which images are formed by the liquid discharge apparatus **1** has reached a predetermined number, or at a timing when the power is turned on to the liquid discharge apparatus **1**, and one or a plurality of semiconductor devices including the control circuit **100** has performed a POR (power on reset).

(62) When the control circuit **100** determines that the number of discharge times of ink onto the medium has reached the predetermined number of times (Y in step S**110**), the control circuit **100** causes the measurement circuit **300** to obtain the capacitance of the piezoelectric element **60** (step S**120**). Specifically, the control circuit **100** outputs an enable signal EN that enables acquisition of the capacitance of the piezoelectric element **60** to the measurement circuit **300**. Thereby, the measurement circuit **300** obtains the capacitance of the piezoelectric element **60** or the physical quantity corresponding to the capacitance of the piezoelectric element **60**.
(63) Next, the determination circuit **310** determines the deterioration state of the piezoelectric element **60** based on the capacitance of the piezoelectric element **60** obtained by the measurement circuit **300** or a physical quantity corresponding to the capacitance of the piezoelectric element **60** (step S**130**). That is to say, the determination circuit **310** determines the deterioration state of the piezoelectric element **60** based on the capacitance of the piezoelectric element **60** measured by the measurement circuit **300**. The life estimation circuit **320** estimates the life of the piezoelectric element **60** based on the determination result of the determination circuit **310** (step S**140**), the correction value calculation circuit **330** calculates the correction value of the drive signal COM (step S**150**), and the control circuit **100** corrects the drive signal COM based on the correction value of the calculated drive signal COM.
(64) The control method of the liquid discharge system SY described above includes a measurement step of measuring the capacitance of the piezoelectric element **60** by the measurement circuit **300**, and a determination step of determining the deterioration state of the piezoelectric element **60** based on the capacitance measured by the determination circuit **310** so that it is possible to detect and determine the deterioration state of the piezoelectric element **60**, which might differ depending on the history of the drive condition, with high accuracy and to improve the detection accuracy and the determination accuracy so that it is possible to perform a suitable correction on the drive condition of the piezoelectric element **60**, and to improve the calculation accuracy of the estimated life of the piezoelectric element **60**.
(65) Here, the drive signal COM is an example of the drive signal, and the drive signal VOUT is also an example of the drive signal in consideration of the point that the drive signal VOUT is a signal generated by determining whether the trapezoid waveform Adp included in the drive signal COM is selected or deselected. Also, the capacitance of the piezoelectric element **60** is an example of the first capacitance, and the capacitance of the dummy piezoelectric element **60***d* is an example of the second capacitance. The correction value calculation circuit **330** that calculates the correction value of the drive signal COM based on the deterioration state of the piezoelectric element **60** and the control circuit **100** that corrects the voltage of the drive signal COM based on the correction value calculated by the correction value calculation circuit **330** represent an example of the drive voltage correction circuit that corrects the voltage of the drive signal COM based on the capacitance of the piezoelectric element **60**.
(66) Operational Advantages
(67) As described above, in the liquid discharge system SY according to the first embodiment, the measurement circuit **300** measures the capacitance of the piezoelectric element **60**, and the determination circuit **310** determines the deterioration state of the piezoelectric element **60** based on the capacitance of the piezoelectric element **60** measured by the measurement circuit **300** so that it is possible to determine deterioration of the amount of change of the piezoelectric element **60** regardless of the drive conditions of the piezoelectric element **60**.
(68) Also, the liquid discharge system SY according to the present embodiment includes the correction value calculation circuit **330** that corrects the voltage of the drive signal COM based on the deterioration state of the piezoelectric element **60** determined by the determination circuit **310**. Thereby, when deterioration occurs in the displacement amount of the piezoelectric element **60**, it is possible to reduce the risk of variation in the discharge amount of liquid, and thus to improve the discharge accuracy of ink in the liquid discharge system SY.

(69) Further, the liquid discharge system SY according to the present embodiment includes the life estimation circuit **320** that estimates the life of the piezoelectric element **60** based on the deterioration state of the piezoelectric element **60** determined by the determination circuit **310** and the notification section **110** that provides a notification of the information on the life of the piezoelectric element **60** estimated by the life estimation circuit **320**. Thereby, it is possible to notify a user of the life of the piezoelectric element **60** and the life of the liquid discharge system SY at a suitable timing, and thus to improve user convenience.
2. Second Embodiment
(70) Next, a description will be given of a liquid discharge system SY according to a second embodiment. In the description of the liquid discharge system SY according to the second embodiment, the same sign is given to the same component as that of the liquid discharge system SY according to the first embodiment, and a detailed description will be simplified or omitted.
(71) FIG. **7** is a diagram illustrating the configuration of the liquid discharge system SY according to the second embodiment. As illustrated in FIG. **7**, the liquid discharge system SY according to the second embodiment includes a dummy piezoelectric element **60***d*. The dummy piezoelectric element **60***d* is not a component included in the discharge section **600** unlike the piezoelectric element **60**. That is to say, the dummy piezoelectric element **60***d* does not have the corresponding vibration plate **621**, cavity **631**, and nozzle **651**. Accordingly, the dummy piezoelectric element **60***d* is displaced in accordance with the drive signal VOUT based on the drive signal COM, but does not discharge ink.
(72) In the liquid discharge system SY according to the second embodiment, the measurement circuit **300** measures the capacitance of the piezoelectric element **60** that discharges ink and the capacitance of the dummy piezoelectric element **60***d* that does not discharge ink. Specifically, the measurement circuit **300** obtains the voltage value of at least one electrode **611** of the n piezoelectric elements **60** as the detection voltage Cc by receiving an input of the enable signal EN that enables detection of a voltage value from the control circuit **100**. The measurement circuit **300** then generates a detection signal Cap corresponding to the capacitance of the piezoelectric element **60** based on the obtained detection voltage Cc and outputs the detection signal Cap to the determination circuit **310**. Also, the measurement circuit **300** obtains the voltage value of one electrode of the dummy piezoelectric element **60***d* as a detection voltage Cd by receiving an input of the enable signal ENd that enables detection of the voltage value from the control circuit **100**. The measurement circuit **300** then generates a detection signal Capd corresponding to the capacitance of the dummy piezoelectric element **60***d* based on the obtained detection voltage Cd and outputs the detection signal Capd to the corresponding determination circuit **310**.
(73) In the liquid discharge system SY according to the second embodiment, the measurement circuit **300** determines the deterioration state of the piezoelectric element **60** based on the capacitance of the piezoelectric element **60** and the capacitance of the dummy piezoelectric element **60***d*. Specifically, the determination circuit **310** compares the detection signal Cap corresponding to the capacitance of the piezoelectric element **60** and the detection signal Capd corresponding to the capacitance of the dummy piezoelectric element **60***d* and determines the deterioration state of the piezoelectric element **60** based on the comparison result. Here, the dummy piezoelectric element **60***d* is not a component included in the discharge section **600** that discharges ink, and thus the stress caused by discharge does not affect the dummy piezoelectric element **60***d*. By using the detection signal Capd indicating the capacitance of the dummy piezoelectric element **60***d* as described above as a reference value and determining the deterioration state of the piezoelectric element **60**, it is possible to determine the deterioration state of the piezoelectric element **60** based on a uniform reference. As a result, it is possible to improve the determination accuracy of the deterioration state by the determination circuit **310**.
(74) Here, it is desirable that the dummy piezoelectric element **60***d* be produced in the same manufacturing process as that of the piezoelectric element **60**. Thereby, the risk of causing

variations in measuring the deterioration state of the piezoelectric element **60**, which are originated from variations in manufacturing, is reduced, and thus it is possible to further improve the determination accuracy of the deterioration state in the determination circuit **310**.
(75) Here, a description will be given of an example of the specific configuration of the measurement circuit **300** according to the second embodiment. FIG. **8** is a diagram illustrating an example of the configuration of the measurement circuit **300** according to the second embodiment. As illustrated in FIG. **8**, the measurement circuit **300** includes the current source **301**, switches **302-1**, **302-2**, **302*d*-1**, and **302*d*-2**, the comparator **303**, and the timer circuit **304**.
(76) The current source **301** is supplied with the voltage Vdd**1**. The current source **301** then generates the constant-value current Icnt based on the voltage Vdd**1**. The current source **301** as described above may be configured, for example, by one or a combination of a plurality of transistors, a constant-current Zener diode, or the like.
(77) One end of the switch **302-1** is electrically coupled to the output of the current source **301**, and the other end is electrically coupled to the electrode **611** of the piezoelectric element **60**. Also, one end of the switch **302-2** is electrically coupled to the electrode **611** of the piezoelectric element **60**, and the other end is electrically coupled to one input end of the comparator **303**. The conduction state of one end and the other end of the switches **302-1** and **302-2** are individually changed based on the enable signal EN input to the respective control ends.
(78) Specifically, when the enable signal EN that enables detection of the capacitance of the piezoelectric element **60** in the measurement circuit **300** is input to the respective control ends of the switches **302-1** and **302-2**, one end and the other end of the switches **302-1** and **302-2** are controlled to be conductive. When the enable signal EN that does not enable detection of the capacitance of the piezoelectric element **60** in the measurement circuit **300** is input to the respective control ends of the switches **302-1** and **302-2**, one end and the other end of the switches **302-1** and **302-2** are controlled to be nonconductive.
(79) One end of the switch **302*d*-1** is electrically coupled to the output of the current source **301**, and the other end is electrically coupled to one electrode of the dummy piezoelectric element **60*d***. Also, one end of the switch **302*d*-2** is electrically coupled to one electrode of the dummy piezoelectric element **60*d***, and the other end is electrically coupled to one input end of the comparator **303**. The conduction state of one end and the other end of the switches **302*d*-1** and **302*d*-2** are individually changed based on the enable signal ENd input to the respective control ends.
(80) Specifically, when the enable signal ENd that enables detection of the capacitance of the dummy piezoelectric element **60*d*** in the measurement circuit **300** is input to the respective control ends of the switches **302*d*-1** and **302*d*-2**, one end and the other end of the switches **302*d*-1** and **302*d*-2** are controlled to be conductive. Whereas when the enable signal ENd that does not enable detection of the capacitance of the dummy piezoelectric element **60*d*** in the measurement circuit **300** is input to the respective control ends of the switches **302*d*-1** and **302*d*-2**, one end and the other end of the switches switch **302*d*-1** and **302*d*-2** are controlled to be nonconductive.
(81) Here, the enable signal EN, which is input to the respective control ends of the switches **302-1** and **302-2**, and the enable signal ENd, which is input to the respective control ends of the switches **302*d*-1** and **302*d*-2**, do not become the logical level that enables detection of the capacitance at the same time.
(82) The voltage Vdd**2** is supplied to the high-potential side power source terminal of the comparator **303**, and the voltage Vss is input to the low-potential side power source terminal of the comparator **303**. Also, one input end of the comparator **303** is electrically coupled to the other end of the switch **302-2** and the other end of the switch **302*d*-2**, and the reference voltage Vref is input to the other input end of the comparator **303**. When the signal input to one input end exceeds the reference voltage Vref, the comparator **303** outputs the H-level output voltage Cp, and when the signal input to one input end is less than or equal to the reference voltage Vref, the comparator **303**

outputs the L-level output voltage Cp.

(83) That is to say, when the enable signal EN that enables detection of the capacitance of the piezoelectric element **60** is input to the measurement circuit **300**, the comparator **303** compares the detection voltage Cc and the reference voltage Vref, and outputs the output voltage Cp based on the comparison result. When the enable signal ENd that enables detection of the capacitance of the dummy piezoelectric element **60***d* is input to the measurement circuit **300**, the comparator **303** compares the detection voltage Cd and the reference voltage Vref, and outputs the output voltage Cp based on the comparison result.

(84) The output voltage Cp output by the comparator **303** is input to the timer circuit **304**. Also, a clock pulse not illustrated in the figure is input to the timer circuit **304** with the enable signals EN and ENd. The timer circuit **304** then measures a time period from when the enable signal EN that enables detection of the capacitance of the piezoelectric element **60** is input to the measurement circuit **300** to when the output voltage Cp output by the comparator **303** becomes H level, and outputs the measurement result as the detection signal Cap. Also, the timer circuit **304** measures a time period from when the enable signal ENd that enables detection of the capacitance of the dummy piezoelectric element **60***d* is input to the measurement circuit **300** to when the output voltage Cp output by the comparator **303** becomes H level, and outputs the measurement result to the determination circuit **310** as the detection signal Capd.

(85) The determination circuit **310** compares the detection signal Cap and the detection signal Capd so as to determine the deterioration state of the piezoelectric element **60**.

(86) As described above, in the liquid discharge system SY according to the second embodiment, in addition to the same operational advantages of the first embodiment, it is possible to determine the deterioration state of the piezoelectric element **60** on a uniform basis by determining the deterioration state of the piezoelectric element **60** by using the detection signal Capd indicating the capacitance of the dummy piezoelectric element **60***d* as a reference value. Accordingly, it is possible to further improve the determination accuracy of the deterioration state of the piezoelectric element **60** in the determination circuit **310**.

3. Third Embodiment

(87) Next, a description will be given of a liquid discharge system SY according to a third embodiment. In the description of the liquid discharge system SY according to the third embodiment, the same sign is given to the same component as that of the liquid discharge system SY according to the first embodiment and the second embodiment, and a detailed description will be simplified or omitted.

(88) FIG. **9** is a diagram illustrating the configuration of a liquid discharge system SY according to the third embodiment. As illustrated in FIG. **9**, the liquid discharge system SY according to the third embodiment includes a liquid discharge apparatus **1** including the piezoelectric element **60** that is displaced and discharges ink based on the drive signal COM, and a server **70** configured to communicate with the liquid discharge apparatus **1**.

(89) The liquid discharge apparatus **1** according to the third embodiment includes a transmission module **350** and a reception module **360** instead of the determination circuit **310**, the life estimation circuit **320**, and the correction value calculation circuit **330**. Also, the server **70** includes a reception module **700**, a transmission module **750**, a determination circuit **710**, a life estimation circuit **720**, and a correction value calculation circuit **730**. Here, the determination circuit **710** held by the server **70** corresponds to the determination circuit **310** according to the first embodiment and the second embodiment, the life estimation circuit **720** held by the server **70** corresponds to the life estimation circuit **320** according to the first embodiment and the second embodiment, and the correction value calculation circuit **730** held by the server **70** corresponds to the correction value calculation circuit **330** according to the first embodiment and the second embodiment.

(90) The transmission module **350** receives the detection signal Cap output by the measurement circuit **300**. The transmission module **350** transmits the input detection signal Cap to the reception

module **700** of the server **70** as a communication signal St. The reception module **700** receives the communication signal St transmitted by the transmission module **350**. The reception module **700** then obtains the detection signal Cap from the received communication signal St and outputs the obtained detection signal Cap to the determination circuit **710** as a detection signal rCap. That is to say, the transmission module **350** and the reception module **700** are configured to communicate with each other. The transmission module **350** and the reception module **700** are connected, for example, by the communication via the internet.
(91) The determination circuit **710** determines the deterioration state of the piezoelectric element **60** held by the liquid discharge apparatus **1** based on the detection signal rCap in the same manner as the determination circuit **310** according to the first embodiment, and outputs a state signal rCst indicating the determination result to the life estimation circuit **720** and the correction value calculation circuit **730**.
(92) The life estimation circuit **720** estimates the life of the piezoelectric element **60** held by the liquid discharge apparatus **1** based on the input state signal rCst in the same manner as the life estimation circuit **320** according to the first embodiment. The life estimation circuit **720** then outputs a life notification signal rEl including the information on the estimated life of the piezoelectric element **60** to the transmission module **750**.
(93) The correction value calculation circuit **730** determines whether or not to correct the voltage of the drive signal COM that drives the piezoelectric element **60** held by the liquid discharge apparatus **1** based on the input state signal rCst in the same manner as the correction value calculation circuit **330** according to the first embodiment. When the correction value calculation circuit **730** determines that the correction of the voltage of the drive signal COM is necessary, the correction value calculation circuit **730** calculates the correction value for correcting the voltage of the drive signal COM and outputs the correction signal rCv including the information on the calculating correction value to the transmission module **750**.
(94) The transmission module **750** transmits a signal including at least one of the life notification signal rEl input from the life estimation circuit **720** and the correction signal rCv input from the correction value calculation circuit **730** to the reception module **360** of the liquid discharge apparatus **1** as a communication signal Sr. The reception module **360** receives the communication signal Sr transmitted from the transmission module **750**. The reception module **360** then generates a state signal Hdi including at least one of the life notification signal rEl and the correction signal Cv from the received communication signal Sr, and outputs the state signal Hdi to the control circuit **100**. That is to say, the transmission module **750** and the reception module **360** are configured to communicate with each other. The transmission module **750** and the reception module **360** are connected, for example, by the communication via the internet.
(95) When the input state signal Hdi includes the life notification signal rEl, and the life notification signal rEl included in the input state signal Hdi includes information indicating that the piezoelectric element **60** will soon reach the end of life, or information indicating that the piezoelectric element **60** has already reached the end of life, the control circuit **100** generates the notification signal Inf for notifying the user that the piezoelectric element **60** will soon reach the end of life or the piezoelectric element **60** has already reached the end of life, and outputs the notification signal Inf to the notification section **110**.
(96) Also, when the input state signal Hdi includes the correction signal Cv, the control circuit **100** corrects the reference drive signal dA in accordance with the input correction signal Cv, and outputs the reference drive signal dA to the drive signal output circuit **50**.
(97) As described above, the liquid discharge system SY according to the third embodiment includes the liquid discharge apparatus **1** including the piezoelectric element **60** and the server **70** configured to communicate with the liquid discharge apparatus **1**. The liquid discharge apparatus **1** includes the piezoelectric element **60** that is displaced in accordance with the drive signal VOUT based on the drive signal COM and discharges ink, the drive signal output circuit **50** that outputs

the drive signal COM, and the measurement circuit **300** that measured the capacitance of the piezoelectric element **60**. The server **70** includes the reception module **700** that receives the communication signal St including the information on the capacitance of the piezoelectric element **60**, the determination circuit **710**, and the transmission module **750** that transmits the communication signal Sr including at least one of the life notification signal rEl in accordance with the state signal rCst, which is the determination result by the determination circuit **710** and the correction signal rCv.
(98) That is to say, the server **70** included in the liquid discharge system SY according to the third embodiment is the server **70** configured to communicate with the liquid discharge apparatus **1** including the piezoelectric element **60** that is displaced in accordance with the drive signal VOUT based on the drive signal COM to discharge ink. The server **70** includes the reception module **700** that receives the communication signal St including the capacitance of the piezoelectric element **60** measured by the measurement circuit **300**, the determination circuit **710** that determines the deterioration state of the piezoelectric element **60** based on the communication signal St, and the transmission module **750** that transmits the communication signal Sr including at least one of the life notification signal rEl in accordance with the state signal rCst, which is the determination result produced by the determination circuit **710**, and the correction signal rCv.
(99) The liquid discharge system SY and the server **70** configured as described above have the same operational advantages as those of the liquid discharge system SY according to the first embodiment.
(100) Here, the communication signal St including the information on the capacitance of the piezoelectric element **60** held by the liquid discharge apparatus **1** is an example of the capacitance information, the reception module **700** that receives the communication signal St is an example of the reception section, the determination circuit **710** of the server **70** is an example of the determination circuit according to the third embodiment, the communication signal Sr based on the state signal rCst indicating the determination result of the deterioration state of the piezoelectric element **60** determined by the determination circuit **710** is an example of the determination result information, and the transmission module **750** that transmits the communication signal Sr is an example of the transmission section.
(101) In the above, descriptions have been given of the embodiments and the variations. However, the present disclosure is not limited to those embodiments. It is possible to carry out the present disclosure in various aspects without departing from the spirit and scope of the disclosure. For example, it is possible to suitably combine the embodiments described above.
(102) The present disclosure includes a component having substantially the same component described in the embodiments (for example, a component having the same function, method and result or a component having the same purpose and advantages). Also, the present disclosure includes a component of which nonessential part is replaced. Also, the present disclosure includes a component configured to have the same operational advantages or accomplish the same purposes as those of the component described in the embodiments. Also, the present disclosure includes a component produced by adding a publicly known technique to the component described in the embodiments.
(103) The following contents are derived from the embodiments described above.
(104) A liquid discharge system including: a piezoelectric element that is displaced based on a drive signal and discharges liquid; a drive signal output circuit that outputs the drive signal; a measurement circuit that measures a first capacitance of the piezoelectric element; and a determination circuit that determines a deterioration state of the piezoelectric element based on the first capacitance.
(105) With the liquid discharge system, the deterioration state of the piezoelectric element is determined based on the capacitance of the piezoelectric element so that it is possible to obtain the displacement amount of the piezoelectric element accurately without depending on the drive

history of the piezoelectric element.

(106) The liquid discharge system described above may further include: a drive voltage correction circuit that corrects a voltage of the drive signal in accordance with the deterioration state.

(107) With the liquid discharge system, it is possible to optimally correct the drive signal in accordance with the displacement amount of the piezoelectric element.

(108) The liquid discharge system described above may further include: a life estimation circuit that estimates a life of the piezoelectric element based on the deterioration state; and a notification section that provides a notification of the information on the life estimated by the life estimation circuit.

(109) With the liquid discharge system, it is possible to notify a user of the life of at least one of the piezoelectric element and the liquid discharge system based on the displacement amount of the piezoelectric element.

(110) In the liquid discharge system described above, the measurement circuit may include a current source that outputs a constant current, a switch that changes electrical coupling with the current source and the piezoelectric element; and a comparator electrically coupled to the piezoelectric element.

(111) With the liquid discharge system, it is possible to measure the capacitance of the piezoelectric element with high accuracy based on the resolution of the clock signal.

(112) The liquid discharge system described above may further include: a dummy piezoelectric element that is displaced in accordance with the drive signal and does not discharge liquid, wherein the measurement circuit may measure a second capacitance of the dummy piezoelectric element, and the determination circuit may determine the deterioration state of the piezoelectric element based on the first capacitance and the second capacitance.

(113) With the liquid discharge system, it is possible to determine the deterioration state of the piezoelectric element in consideration of the parameters, such as temperature, humidity, or the like.

(114) The liquid discharge system described above may further include: a liquid discharge apparatus including the piezoelectric element; and a server configured to communicate with the liquid discharge apparatus, wherein the server may include a reception section that receives capacitance information including the first capacitance, the determination circuit, and a transmission section that transmits determination result information including a determination result of the determination circuit.

(115) A server configured to communicate with a liquid discharge apparatus including a piezoelectric element that is displaced based on a drive signal and discharges liquid, the server may include: a reception section that receives capacitance information including a capacitance of the piezoelectric element measured by a measurement circuit; a determination circuit that determines a deterioration state of the piezoelectric element based on the capacitance information; and a transmission section that transmits determination result information including a determination result of the determination circuit.

(116) With the server, the deterioration state of the piezoelectric element is determined based on the capacitance of the piezoelectric element so that it is possible to obtain the displacement amount of the piezoelectric element accurately without depending on the drive history of the piezoelectric element.

(117) A control method of a liquid discharge system including a piezoelectric element that is displaced based on a drive signal and discharges liquid, and a drive signal output circuit that outputs the drive signal, the control method may include: a measurement step of measuring a capacitance of the piezoelectric element; and a determination step of determining a deterioration state of the piezoelectric element based on the capacitance.

(118) With the liquid discharge system, the deterioration state of the piezoelectric element is determined based on the capacitance of the piezoelectric element so that it is possible to obtain the

displacement amount of the piezoelectric element accurately without depending on the drive history of the piezoelectric element.

Claims

1. A liquid discharge system comprising: a piezoelectric element that is displaced based on a drive signal and discharges liquid; a drive signal output circuit that outputs the drive signal; a measurement circuit that measures a first capacitance of the piezoelectric element; a determination circuit that determines a deterioration state of the piezoelectric element based on the first capacitance; a life estimation circuit that estimates a life of the piezoelectric element based on the deterioration state and generates a life notification signal including information on an estimated life of the piezoelectric element to notify a user that the piezoelectric element will reach an end of the life of the piezoelectric element, or that the piezoelectric element has already reached the end of the life of the piezoelectric element; a control circuit that generates, based on the life notification signal, a notification signal that notifies the user that the piezoelectric element will reach the end of the life of the piezoelectric element, or that the piezoelectric element has already reached the end of the life of the piezoelectric element, and that outputs the notification signal; and a notification section that provides the user with a notification of information on the life of the piezoelectric element based on the notification signal.
2. The liquid discharge system according to claim 1, further comprising: a drive voltage correction circuit that corrects a voltage of the drive signal in accordance with the deterioration state.
3. The liquid discharge system according to claim 1, wherein the measurement circuit includes a current source that outputs a constant current, a switch that changes electrical coupling with the current source and the piezoelectric element; and a comparator electrically coupled to the piezoelectric element.
4. The liquid discharge system according to claim 1, further comprising: a dummy piezoelectric element that is displaced in accordance with the drive signal, but does not discharge liquid, wherein the measurement circuit measures a second capacitance of the dummy piezoelectric element, and the determination circuit determines the deterioration state of the piezoelectric element based on the first capacitance and the second capacitance.
5. The liquid discharge system according to claim 1, further comprising: a liquid discharge apparatus including the piezoelectric element; and a server configured to communicate with the liquid discharge apparatus, wherein the server includes a reception section that receives capacitance information including the first capacitance, the determination circuit, and a transmission section that transmits determination result information including a determination result of the determination circuit.
6. A server configured to communicate with a liquid discharge apparatus including a piezoelectric element that is displaced based on a drive signal and discharges liquid, the server comprising: a reception section that receives capacitance information including a capacitance of the piezoelectric element measured by a measurement circuit; a determination circuit that determines a deterioration state of the piezoelectric element based on the capacitance information; a life estimation circuit that estimates a life of the piezoelectric element based on the deterioration state and generates a life notification signal including information on an estimated life of the piezoelectric element to notify a user that the piezoelectric element will reach an end of the life of the piezoelectric element, or that the piezoelectric element has already reached the end of the life of the piezoelectric element; and a transmission section that transmits the life notification signal.
7. A control method of a liquid discharge system including a piezoelectric element that is displaced based on a drive signal and discharges liquid, and a drive signal output circuit that outputs the drive signal, the control method comprising: measuring a capacitance of the piezoelectric element; determining a deterioration state of the piezoelectric element based on the capacitance; estimating a

life of the piezoelectric element based on the deterioration state, and generating a life notification signal including information on an estimated life of the piezoelectric element to notify a user that the piezoelectric element will reach an end of the life of the piezoelectric element, or that the piezoelectric element has already reached the end of the life of the piezoelectric element; generating, based on the life notification signal, a notification signal that notifies the user that the piezoelectric element will reach the end of the life of the piezoelectric element, or that the piezoelectric element has already reached the end of the life of the piezoelectric element, and outputting the notification signal; and providing the user with a notification of information on the life of the piezoelectric element based on the notification signal.